Sanjay Sakthirajan

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Summary

M.S. candidate in Data Science for Operations Research at the University of Minnesota with experience in data analytics, machine learning, and AI-powered applications. Skilled in Python, SQL, statistical analysis, and building end-to-end data solutions using machine learning and LLMs. Experienced in analyzing operational data and process performance metrics to identify bottlenecks and support optimization initiatives, contributing to an increase in production throughput from 20 to 22 units per shift. Built AI- and data-driven projects including an LLM-powered invoice extraction system, predictive machine learning models, and recommendation systems to generate actionable insights and support data-driven decision-making.

Employment history

Ganga Associates **Jun 2022 - Sep 2022**

Operations Intern *Trichy, India*

- Analyzed production and operational data through time studies and workflow analysis, contributing to process improvements that increased line output from 20 to 22 units per shift.
- Performed workflow and capacity analysis to identify production bottlenecks and recommend process improvements.
- Collected and analyzed production performance metrics to support operational decision-making and continuous improvement.
- Collaborated with production and engineering teams to investigate process issues and support process optimization initiatives.

PSG Industries **Aug 2020 - Jun 2024**

Industrial Trainee *Coimbatore, India*

- Collected, organized, and analyzed manufacturing performance data using Microsoft Excel and time studies to identify opportunities for process efficiency improvements.
- Evaluated workflow and process performance metrics to identify inefficiencies and support continuous improvement initiatives.
- Developed process maps and operational documentation to improve visibility into manufacturing workflows.
- Collaborated with production teams to monitor manufacturing performance trends and identify opportunities for process optimization.

Skills

SQL Databases, Machine Learning, Python, R, Matlab, pandas, Numpy, Matplotlib, Scikit-Learn, Data Cleaning, Exploratory Data Analysis, Statistical Analysis, Regression Analysis, Power BI, Microsoft Excel, Git, GitHub, Jupyter Notebook, PyCharm, AWS

Education

Master's, Data Science for operations Research **Sep 2025 - Dec 2026**

University of Minnesota-Twin Cities *Minneapolis, Minnesota*

- **GPA:** 3.87/4.00
- **Coursework:** Data Science for Operations Research, Regression Analysis, Statistical Analysis, Optimization, Quality and Reliability, Project Management, Machine Learning, Analytics for Data Driven Decision Making, Decision Analysis, Game Theory

Bachelor's, Mechanical Engineering **Jul 2020 - Jul 2025**

PSG College of Technology *Coimbatore*

- **GPA:** 3.17/4.00
- **Coursework:** Operations Research, Probability and Statistics, Manufacturing Systems, Quality Engineering, Supply Chain Management, Production Planning and Control, Engineering Economics

Certifications

- AWS Cloud Practitioner: AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services | Jul 2026

Custom section

Invoice Data Extractor

- Built an end-to-end AI-powered invoice extraction system using Claude API, Python, Pydantic, and MySQL to convert unstructured PDF invoices into structured data.
- Designed schema-based information extraction with Pydantic models to accurately extract invoice details, vendor information, and line items from diverse invoice formats.
- Developed a relational MySQL database to store extracted invoices and associated line items, enabling efficient querying and structured data management.
- Built and deployed an interactive Streamlit application for invoice upload, automated extraction, database integration, and real-time results visualization.
- Evaluated extraction quality using field-level accuracy and consistency validation to assess the reliability of LLM-generated outputs.

NBA Championship Predictor

- Designed an end-to-end predictive analytics pipeline by collecting 25 seasons of NBA team statistics using the NBA API and storing the data in a MySQL database.
- Engineered advanced basketball performance metrics and trained machine learning models using scikit-learn to predict NBA championship probabilities.
- Performed feature engineering and model evaluation to identify the statistical factors most strongly associated with championship success.
- Developed and deployed an interactive Streamlit application that enables users to explore championship predictions and team performance metrics in real time.

Netflix Recommendation App

- Developed a content-based movie recommendation system using TF-IDF, cosine similarity, and scikit-learn to generate personalized movie recommendations.
- Performed data preprocessing, feature engineering, and text vectorization on movie metadata to improve recommendation quality and similarity matching.
- Utilized Python and pandas to clean, transform, and analyze movie datasets for content-based recommendation modeling.
- Developed and deployed an interactive Streamlit application that delivers real-time movie recommendations based on user-selected titles.